MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL MX-9872(V)6/UA

This specification sheet is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)6/UA Scanner shall consist of this document and MIL-S-49166.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-9872(V)6/UA and augments MIL-S-49166. The MX-9872(V)6/UA is used on the Bradley Fighting Vehicle, Integrated Sight Unit.

1.2 Purpose. The purpose of this specification sheet is to modify or expand the contents of MIL-S-49166 to describe the MX-9872(V)6/UA Mechanical Scanner. All the MIL-S-49166 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49166 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49166 Scanner, Mechanical, MX-9872(V)/UA, General Specification for

DRAWINGS

USA ELECTRONICS R&D COMMAND DRAWINGS

80063	DLSM-B-771731	Scan-Interlace, PL-1403/UA

USA TANK AUTOMOTIVE COMMAND DRAWINGS

19207	12293969	Scanner Assembly, Mechanical

3. REQUIREMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.1 Description. The MX-9872(V)6/UA Scanner, herein known as the MX-9872(V)6/UA, shall consist of the MIL-S-49166, MX-9872(V)/UA Scanner with a null lens and counterweight assembled and adjusted for 30 Hz operation.

3.2 Construction. The MX-9872(V)6/UA shall be constructed in accordance with drawing 19207 12293969.

3.2.1 Weight. Weight of the MX-9872(V)6/UA shall be 1.75 pounds maximum.

3.5 Components.

a. Scanner Mechanical MX-9872(V)/UA
b. Null lens
c. Counterweight

3.6 Performance characteristics. The characteristics specified herein shall apply as defined in 3.1 and 3.6.1 except 4.7.4, and 4.7.5 apply with the null lens, counterweight and associated hardware removed. Operation shall be at 30 Hz using the Scan-Interlace, PL-1403/UA (DLSM-B-771731).

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (30 Hz). Scan frequency of the MX-9872(V)6/UA with the Scan-Interlace (DLSM-B-771731) shall be adjusted to 30.0, ±0.5 Hz. Scan frequency drift shall not be greater than ±1.5 Hz from the frequency at ambient setting (see MIL-S-49166, 4.4) when exposed to high and low temperature environmental conditions (see MIL-S-49166).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be 2.75, ±0.10 degrees in each direction from a true 45-degree position with respect to the mounting surface (Datum -A- in figure 4) required for securing the MX-9872(V)6/UA.

3.6.2 Interlace. The MX-9872(V)6/UA shall operate in the 2:1 interlace mode. The interlace switching time shown in MIL-S-49166, figure 1, shall not be greater than 4.0 milliseconds (ms).

3.6.3 Scan direction. The MX-9872(V)6/UA shall show the waveform shapes and phase relationships in figure 1 while operaing in either the 2:1 side or the 2:1 rear interlace mode.

3.6.4 Power consumption. Power required to operate the MX-9872(V)6/UA shall not exceed 2.2 watts.

3.6.5 Scan efficiency. The MX-9872(V)6/UA scan efficiency shall be a minimum of 75 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 62.0 microseconds maximum.

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3.7 Environmental conditions. See MIL-S-49166 except as follows:

3.7.4 Shock. See MIL-S-49166 and 3.7.4.1.

3.7.4.1 Shock and scan frequency. Scan frequency (3.6.1.2) shall return to normal within 0.2 second after the MX-9872(V)6/UA is subjected to the operating shock environment.

3.7.5 Vibration. The MX-9872(V)6/UA shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 2 for the X axis and as shown in figure 3 of MIL-S-49166 for the Y and Z axis. During specified vibration, the average interlace waveform amplitude shown in MIL-S-49166, figure 1 shall not deviate from the amplitude without vibration by more than ±50 percent.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the DLSM-B-771731 Scan-Interlace with the MX-9872(V)6/UA with a scan angle of 5.5, ±0.2 degrees and at the 30.0, ±0.5 Hz scan rate.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

4.6.1.2 Specific scan frequency (30 Hz). The MX-9872(V)6/UA shall be connected to the Scan-Interlace, DLSM-B-771731, and mounted to a suitable fixture. Power shall be applied and the output of the tachometer (Scan-Interlace DLSM-B-771731) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (DLSM-B-771731) shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

4.6.1.3 Synchronous scan frequency. Not applicable.

4.6.1.4 Scan angle.

4.6.1.4.1 Test apparatus. The test apparatus shall consist of a holding fixture utilizing the mounting surface (Datum -A- in figure 4) required for securing the MX-9872(V)6/UA, a projection system capable of projecting a spot or marker into the scanner along the optical axis (figure 4) and an appropriate scale graduated in degrees measured about the scan mirror axis. The zero position of this scale shall be the position where the spot or marker is projected onto the scale when the mirror is at a true 45 degrees with respect to the mounting surface (Datum -A- in figure 4) required for securing the MX-9872(V)6/UA.

4.6.1.4.2 Test conduction. The scanner shall be operated and the angular excursions of the projected spot in each direction shall be recorded.

4.6.1.4.3 Failure criteria. Failure to meet the requirement of 3.6.1.4 shall constitute failure of this test.

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4.6.2 Interlace. See MIL-S-49166.

4.6.3 Scan direction. With the MX-9872(V)6/UA operating in either the 2:1 side or the 2:1 rear interlace mode, the outputs of the interlace transducer and the scan tachometer (Scan-Interlace DLSM-B-771731) are observed with an oscilloscope. The output waveforms shall show the applicable shape and phase relationships shown in figure 1. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49166.

4.6.5 Scan efficiency. The MX-9872(V)6/UA shall be operated with the Scan-Interlace (DLSM-B-771731) and measurements taken as follows: The period of the scan tachometer is measured by an oscilloscope. The active time between two consecutive scans shall be determined by measuring the time between the 10 percent and the 90 percent points of the scan tachometer waveform. Scan efficiency shall be calculated by dividing the sum of the two consecutive scans by the scan tachometer period and multiplying by 100 percent. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Scan jitter in azimuth. The MX-9872(V)6/UA shall be tested as specified in MIL-S-49166 with additional circuitry used as shown in figure 3 to test the apparatus. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.7 Environmental tests.

4.7.4 Shock. See 4.7.4.1 and MIL-S-49166.

4.7.4.1 Shock and scan frequency. Failure to meet the requirements of 3.7.4.1 during the operating test shall constitute failure of this test.

4.7.5 Vibration. See figure 2 and MIL-S-49166.

6. NOTES

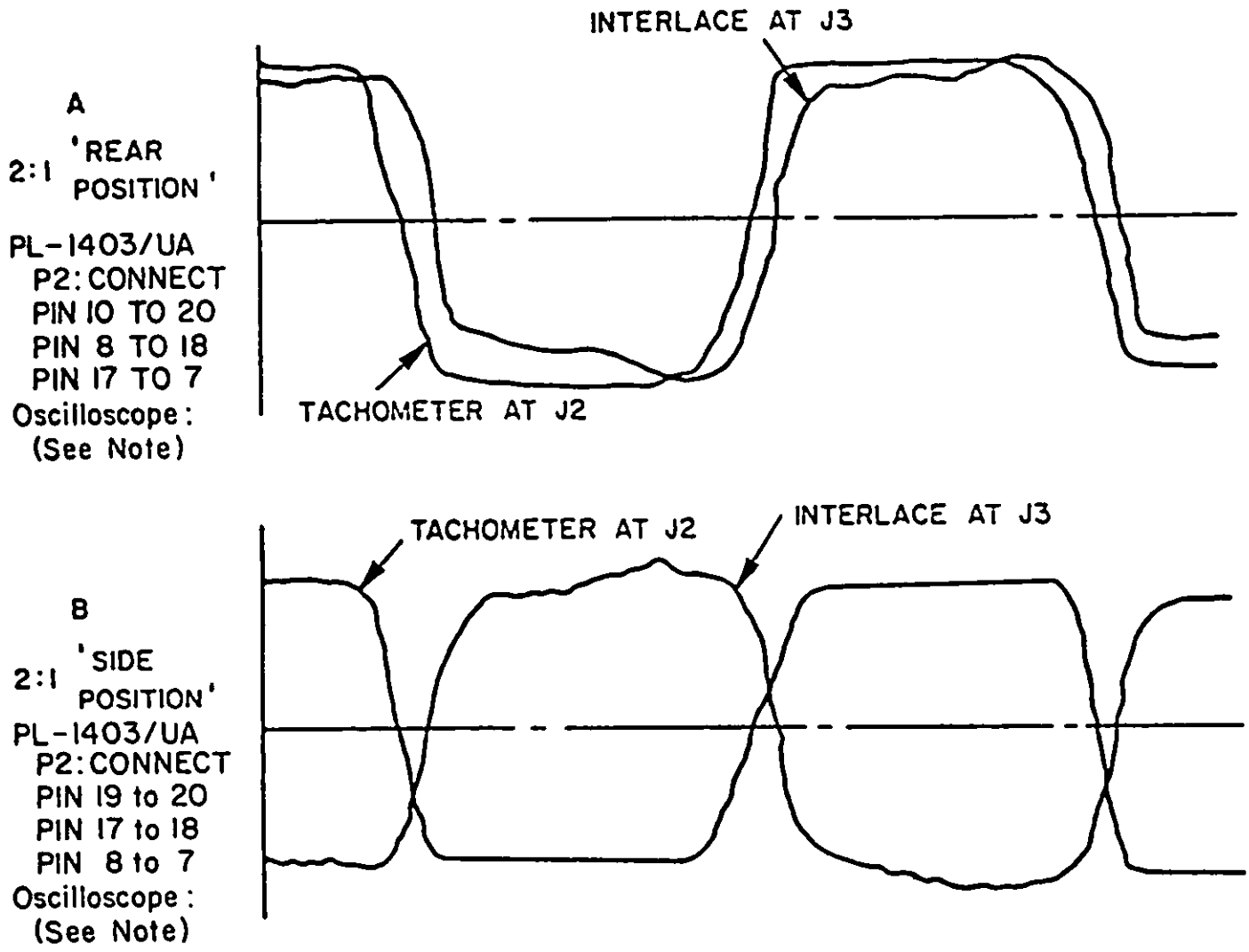
6.1 Intended use. The MX-9872(V)6/UA is intended for use with the Bradley Fighting Vehicle, Integrated Sight Unit.

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NOTE - CONNECT OSCILLOSCOPE AS FOLLOWS:

CHANNEL 1 - (TACHOMETER) HIGH INPUT TO J2 (OR J1-13) AND LOW SCOPE TO GND.
CHANNEL 2 - (INTERLACE) HIGH INPUT TO J3 (OR J1-1) AND LOW SCOPE TO GND.

FIGURE 1. SHAPE AND PHASE RELATIONSHIP OF INTERLACE TRANSDUCER AND SCAN TACHOMETER.

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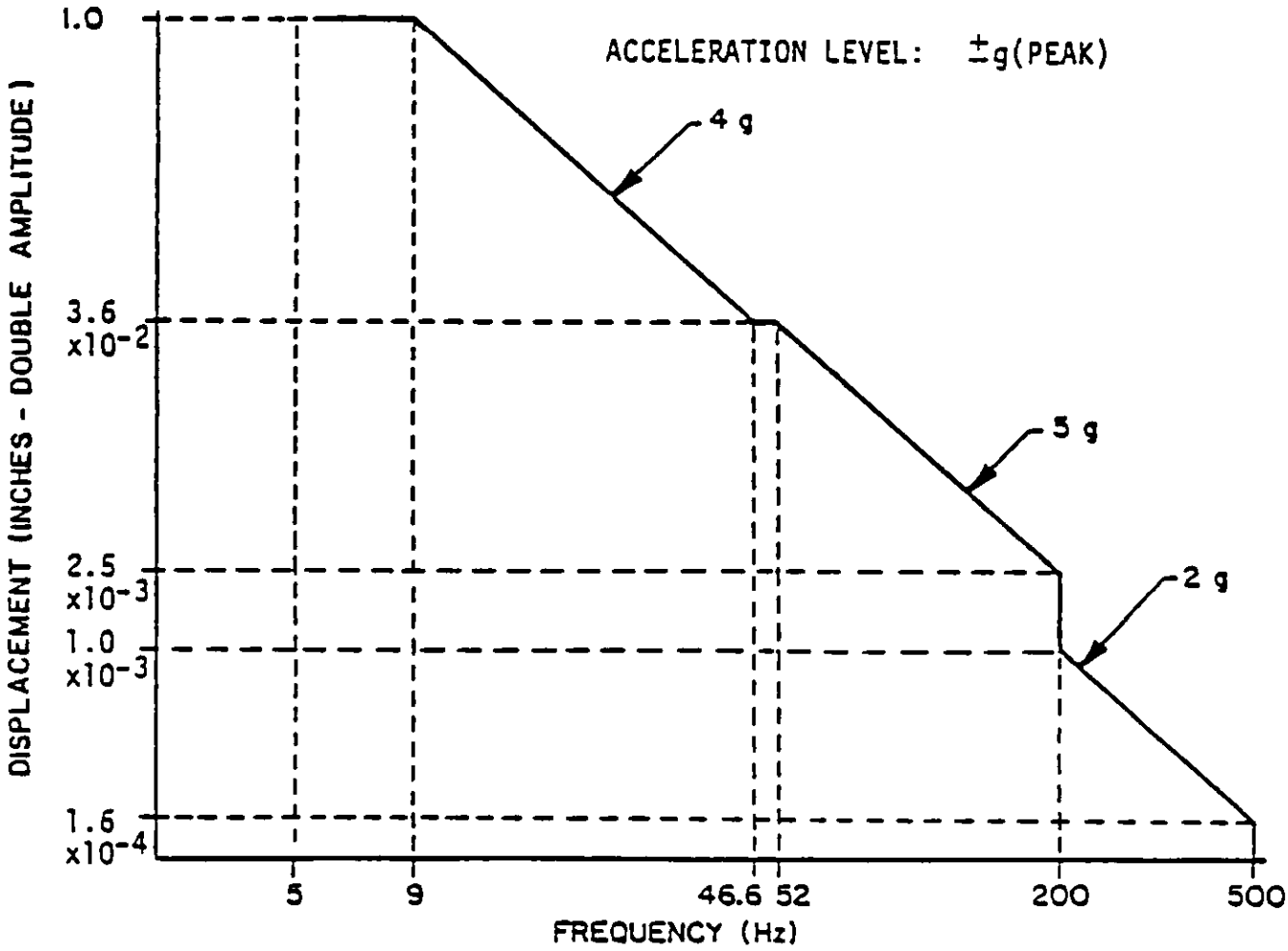


FIGURE 2. VIBRATION TEST PROFILE (X AXIS).

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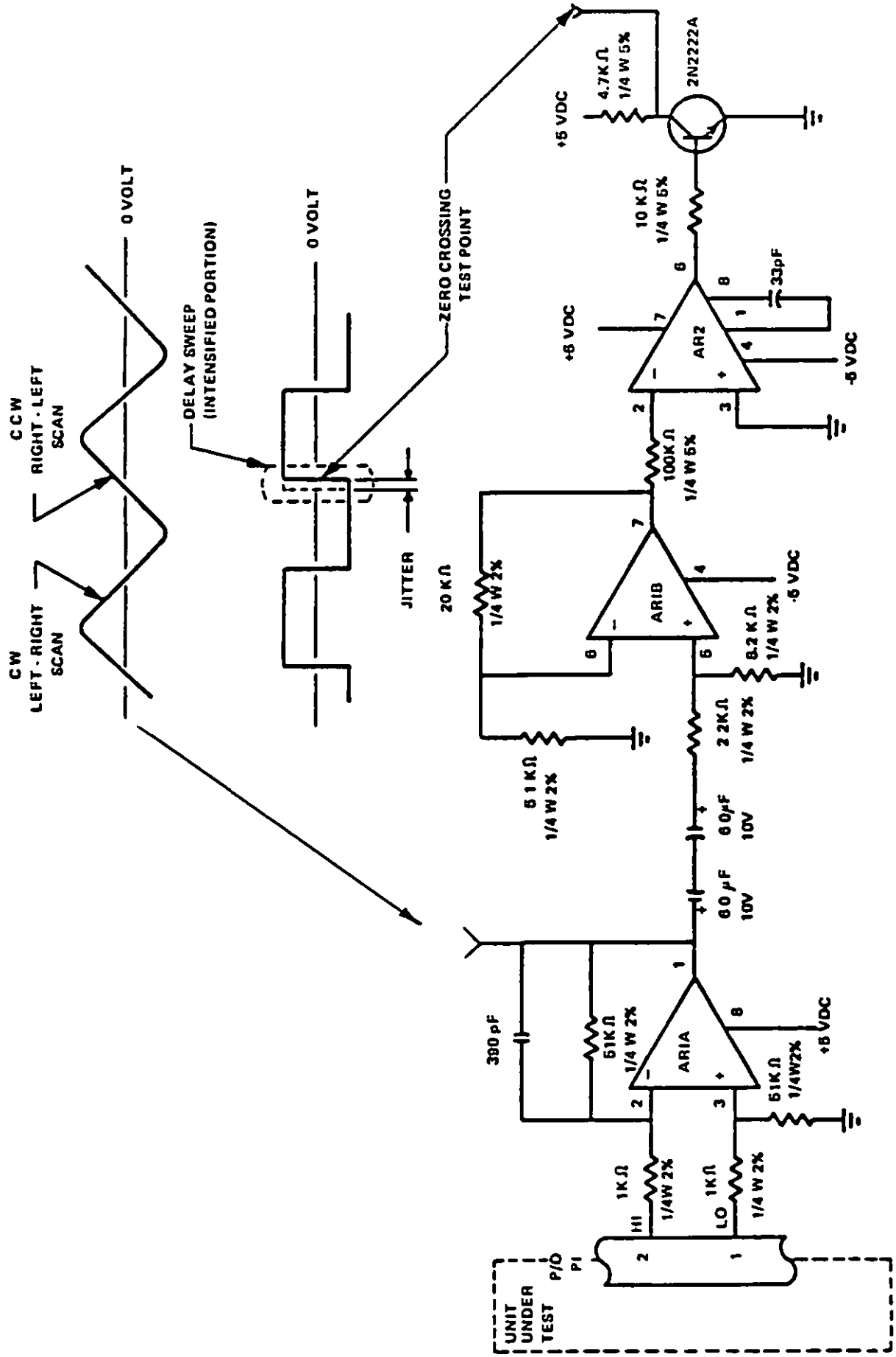
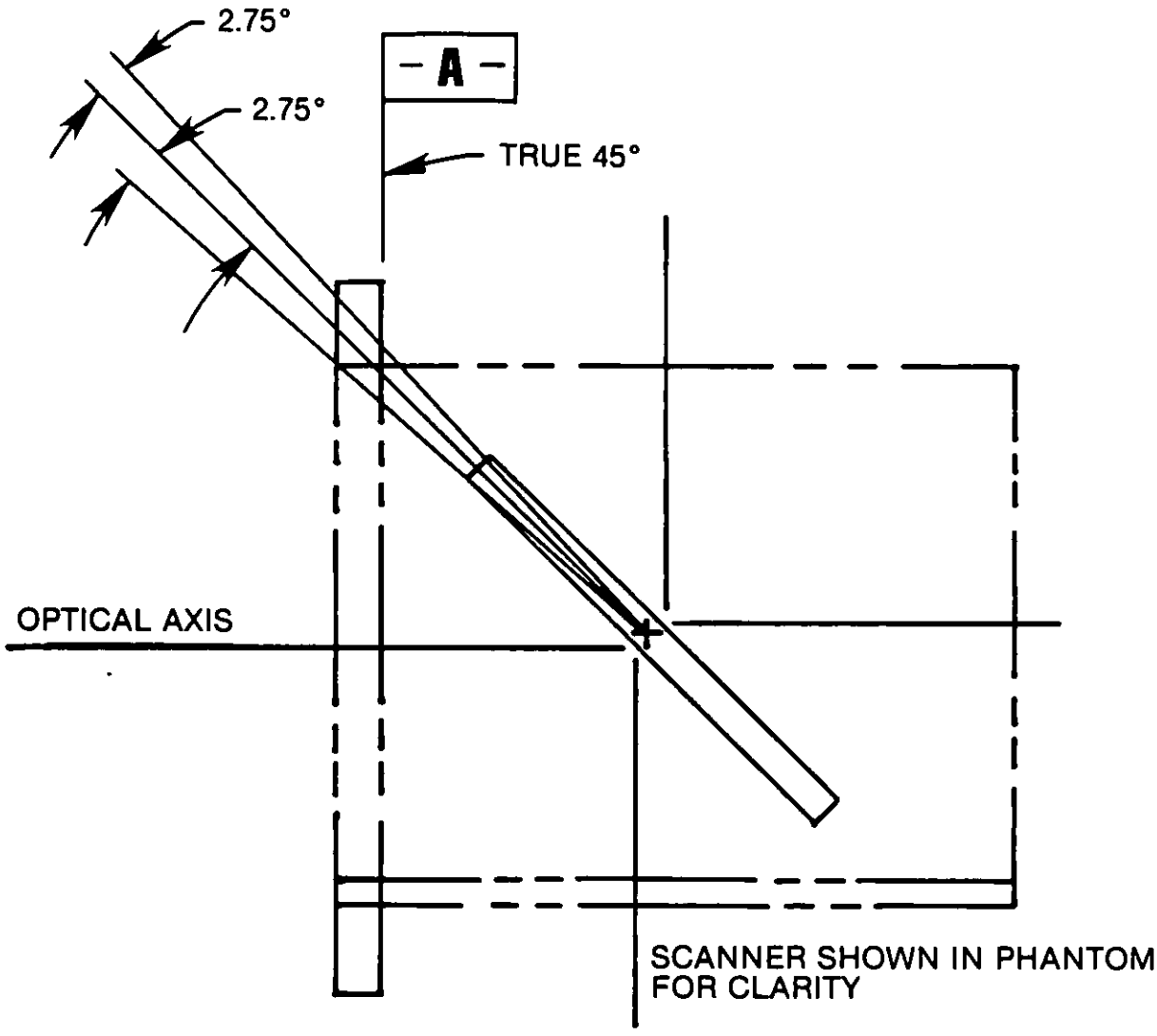


FIGURE 3. TYPICAL CIRCUITRY FOR SCAN JITTER TEST.

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THE OPTICAL AXIS IS DEFINED TO BE ORTHOGONAL TO THE MOUNTING SURFACE, DATUM -A-.

Figure 4. SCAN ANGLE.

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